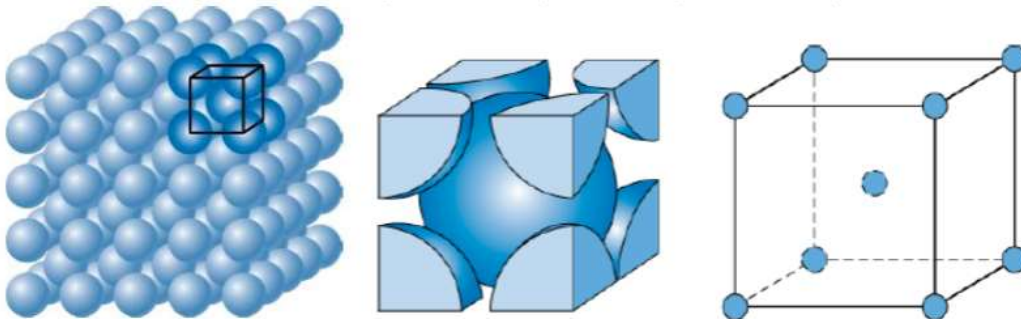


Crystal structure

Body centered cubic



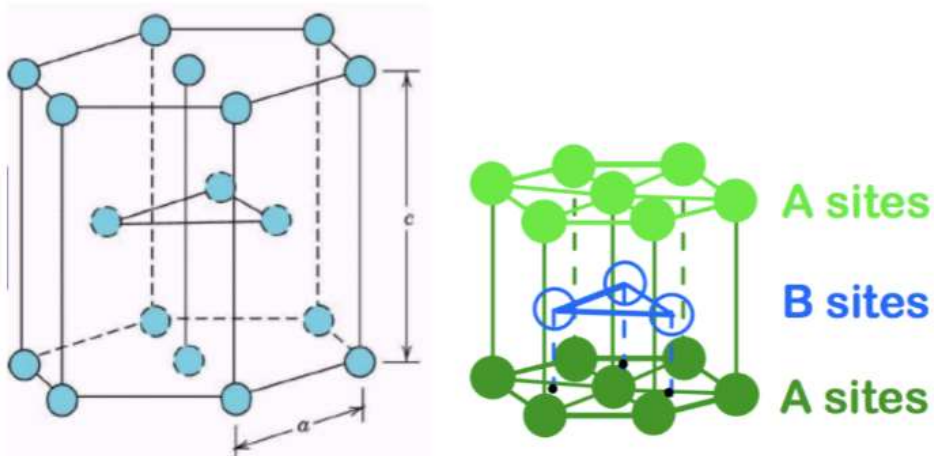
2 atom per unit cell ($1/8$ in each corner + 1)

Coordination # : 8

APF : 0.68 (atoms touch along the cube diagonal)

Hexagonal Close Packed (HCP)

- Coordination # 12
- Number of atoms per unit cell: 12 (corner atoms) $\times \frac{1}{6} + 3$ (interior atoms) + 2 (face atoms) $\times \frac{1}{2} = 6$ atoms / unit cell
- $c/a = 1.63$ For most HCP metals

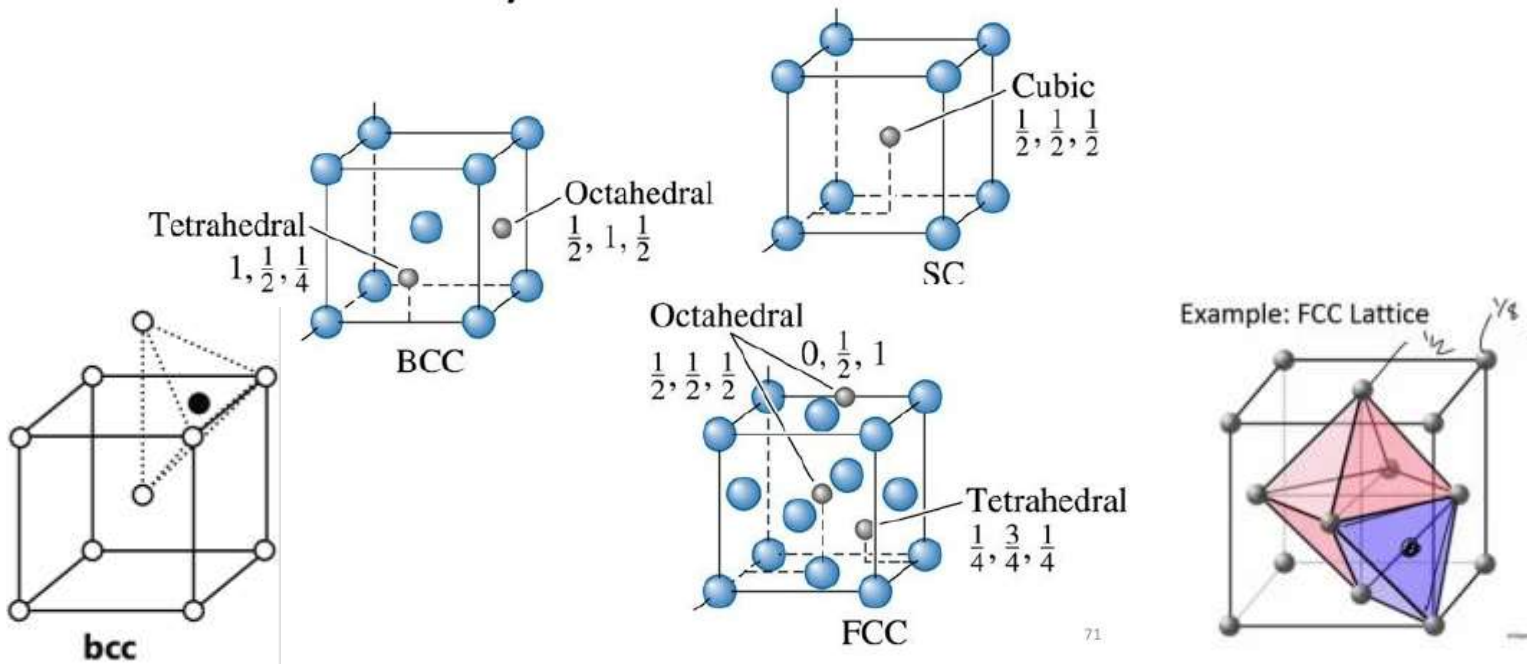


Explain how FCC and HCP are the same (crystal orientation)

Explain the difference between ABABAB and ABCABC stackings

Interstitial sites

Interstitial sites / voids

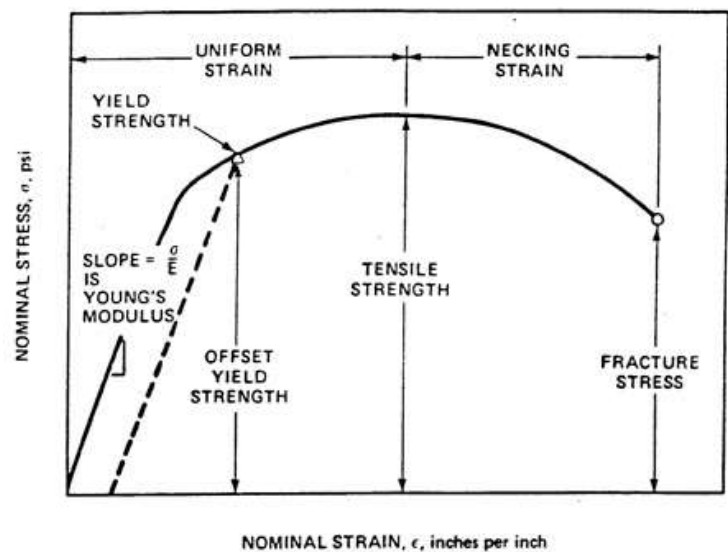


Strain-Stress curve for a typical metal

Identify every point on the curve : elastic proportional (linear) limit, yield strength (elastic limit), ultimate tensile strength, fracture strength, linear elastic region.

Explain reasoning behind the 0.2% offset for the yield strength

0.2% offset is selected large enough to show start of plastic deformation and it cause small amount of def. so if we load sample to this point, atoms will move to new equilibrium position



Learning objectives of the week (accessible in the syllabus section in Quercus)

Make sure you're familiar with these concepts for your tutorials

-
- Calculate the theoretical density of a BCC metal given the atomic weight, crystal structure and atomic radius
 - Demonstrate that the coordination number for BCC is 8
 - Show that the atomic packing factor (APF) for BCC is 0.68
-
- Calculate the geometrically ideal interstitial site size for the following interstitial sites: octahedral, simple cubic
 - Using the hexagonal close packed crystal structure, illustrate how the theoretical density of a polycrystalline metal can be calculated from knowledge of the crystal structure, atomic radius, and atomic weight
 - Calculate the theoretical density of an HCP metal given the atomic weight, crystal structure and atomic radius
 - Demonstrate that the coordination number for HCP is 12
 - Show that the atomic packing factor (APF) for FCC is 0.74
 - Demonstrate how the FCC and HCP crystal structures may be formed from stacking close packed planes in an ABCABC or an ABAB pattern
-
- Identify on the stress-strain curve for a typical metal the following points or regions: proportional limit, yield strength, ultimate tensile strength, fracture strength, linear elastic region
 - Explain the rationale for the use of the 0.2% offset yield strength
-

Questions

1. Which of the following statements is TRUE?

- a) The coordination number of an octahedral hole or interstitial site is 8.
- b) In a rock salt crystal, atoms touch along the unit cell edge.
- c) A rock salt crystal structure can be described as an FCC array of an ions with cations occupying the tetrahedral holes or interstitial sites.
- d) A ceramic with a general chemical formula of AX_2 may have the rock salt crystal structure.

Answer: B

2. Which of the following is equal to the volume of the BCC unit cell in terms of the atomic radius R:

- a) $8R^3$
- b) $22.63 R^3$
- c) $11.31 R^3$
- ☒ d) $12.32 R^3$

Answer: ~~A~~

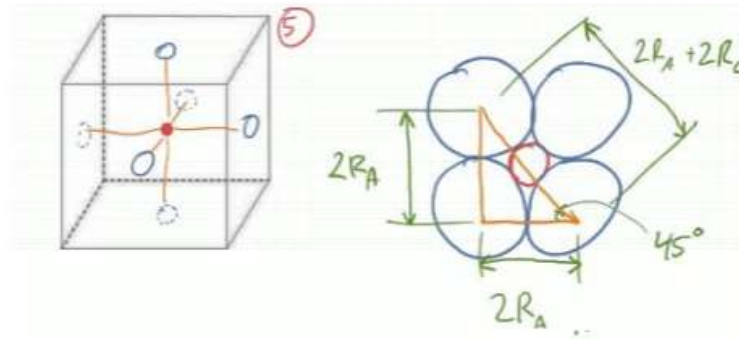
3. For the interstitial site having coordination number of 8, the cation sits at the center of a _____, where the corners are occupied by anions.

- a) Triangle
- b) Octahedron
- ☒ c) Cube
- d) Tetrahedron

Longer question

I. Demonstrate that the ideal cation-to-anion ratio (R_C/R_A) for a cation that just fits into the octahedral interstitial site, without pushing the anions apart is 0.414. Begin with a sketch of one octahedral site within the unit cell below, clearly showing the coordination number.

(see solution in tutorial 3)



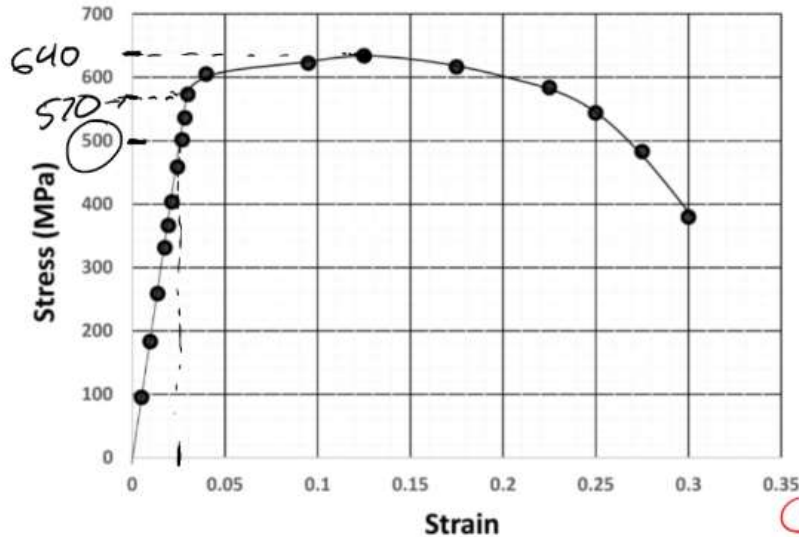
$$\sin 45 = \frac{2R_A}{2R_A + 2R_C}$$

$$R_A \sin 45 + R_C \sin 45 = R_A$$

$$\frac{R_A}{R_A} \sin 45 = \frac{R_C}{R_A} \sin 45 = \frac{R_A}{R_A}$$

$$\frac{R_C}{R_A} = \frac{1 - \sin 45}{\sin 45} = 0.414$$

2. You are evaluating a hypothetical metal alloy specimen with a cross-sectional area of 0.7 cm^2 and length of 10 cm . The tensile stress-strain behavior of this alloy is provided below.



- a. What is the Young's modulus of this specimen?

$$E = \frac{\sigma}{\epsilon} = \frac{570(10^6) \text{ Pa}}{0.025} = 2(10^{10}) \text{ Pa} = 20 \text{ GPa}$$

accept $\sim 18 - 22 \text{ GPa}$

- b. What is the maximum load (in N) that can be sustained by the specimen?

$$\sigma = \frac{F}{A} \Rightarrow F = \sigma \cdot A = 640(10^6) \cdot 0.7 \text{ cm}^2 \left(\frac{\text{m}}{10^2 \text{ cm}} \right)^2 = 44.8 \text{ kN}$$

$\sim 45 \text{ kN}$ (circled 2) $\sim 42-48 \text{ kN}$

- c. What is the maximum load (in N) that can be applied to the specimen without plastic deformation?

$$F = 570(10^6) \cdot 0.7 \cdot \frac{1}{10^4} = 39.9 \text{ kN}$$

$\sim 40 \text{ kN}$ (circled 2) (circled 2)

$\sim 38-42 \text{ okay}$

4. The volume of an HCP unit cell is $\frac{3\sqrt{3}}{2}a^2c$. Show that the atomic packing factor of the HCP crystal structure is 0.74. Begin by sketching the atom positions, as small dots, in the unit cell below. (10)

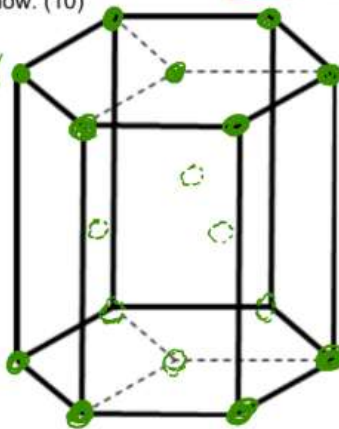
Since $APF = 0.74$.



$\frac{1}{6}$ at each vertex

$\frac{1}{2}$ each face

3 internal



and $a = 2R$.

or $R = \frac{a}{2}$

atom positions (3)

$$\text{so } 12 \cdot \frac{1}{6} + 2 \cdot \frac{1}{2} + 3 = 6 \text{ atoms.}$$

$$0.74 = \frac{6 \cdot \frac{4}{3} \pi R^3}{\frac{3\sqrt{3}}{2} (2R)^2 \cdot c}$$

$$\Rightarrow c = \frac{2 \cdot 6 \cdot 4 \cdot \pi \cdot a^3}{2 \cdot 3 \cdot 3 \cdot \sqrt{3} \cdot 4 \cdot 0.74}$$

$$\frac{c}{a} = \frac{2\pi}{3\sqrt{3} \cdot 0.74}$$

okay if memorized this.

$$= 1.63 \Rightarrow c = 1.63 \cdot a = 1.63 \cdot (2R)$$

$$\text{so, } APF = \frac{6 \cdot \frac{4}{3} \pi R^3}{\frac{3\sqrt{3}}{2} (2R)^2 \cdot 1.63(2R)} = \frac{6 \cdot 4 \cdot \pi \cdot 2}{3\sqrt{3} \cdot 4 \cdot 2 \cdot 1.63}$$

$$= \frac{2\pi}{3\sqrt{3} \cdot 1.63} = 0.74$$

work (4)

or (1) knowing

$a = 2R$

(1) for $c = 3.26R$